QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/483A(USAF)
15 June 1984
SUPERSEDING
MIL-M-38510/483(USAF)
31 August 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE

MONOLITHIC COUNTER/TIMER UNIT

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by Rome Air Development Center, Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic, N-channel, silicon gate counter/timer unit. One product assurance class and a choice of lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Frequency	Circuit
01	4.0 MHz	Counter/timer unit
02	2.5 MHz	Counter/timer unit

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
X	See figure 1 (28 lead, 9/16" x 1 7/16"), dual-in-line package

1.3 Absolute maximum ratings.

V_{CC} supply voltage range (referenced to ground)	-0.3 to +7 V dc
Voltage on any pin (referenced to ground)	-0.3 to +7 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation:	
At -55°C	0.7 W
At +125°C	0.5 W
Lead temperature (soldering 10 seconds)	+270°C
Maximum junction temperature (T_J)	
at T_C = 125°C	+180°C
Thermal resistance, junction to case (θ_{JC})	48°C/W

1.4 Recommended operation conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH}):	
Logic inputs	2.0 V dc
Clock input	V_{CC} - 0.6 V dc
Maximum low level input voltage (V_{IL}):	
Logic inputs	0.8 V dc
Clock input	0.45 V dc

Benefical comments (recommendations, additions, deletions) and any pertinent data which any be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Frequency of operation:	
01 circuit- - - - - - - - - - - - - - - - - - - -	0.5 to 4.0 MHz
02 circuit- - - - - - - - - - - - - - - - - - - -	0.5 to 2.5 MHz
Case operating temperature range (T_C) - - - - - -	-55°C to +125°C
Clock rise time:	
01 circuit- - - - - - - - - - - - - - - - - - - -	20 ns maximum
02 circuit- - - - - - - - - - - - - - - - - - - -	30 ns maximum
Clock fall time:	
01 circuit- - - - - - - - - - - - - - - - - - - -	20 ns maximum
02 circuit- - - - - - - - - - - - - - - - - - - -	30 ns maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification and standard, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 2.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 3.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.5.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.6).

		PAGE
TABLE I.	Electrical performance characteristics, dc and ac - - - - -	3
II.	Electrical test requirments - - - - - - - - - - - - - - - -	6
III.	Group A inspection- -	21
IV.	Input/output conditions for table III - - - - - - - - - - -	24
V.	Register definitions- - - - - - - - - - - - - - - - - - - -	28
VI.	Test vectors (see 6.1)- - - - - - - - - - - - - - - - - - -	34

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TABLE I. Electrical performance characteristics.

Parameter	Symbol	Conditions $-55 \leq T_C \leq +125^{\circ}C$	Min	Max	Unit
Clock high input voltage	V_{IH1}		V_{CC}-0.6	5.5	V
Clock low input voltage	V_{IL1}		-0.3	0.45	V
Logic input high voltage	V_{IH2}		2.0		V
Logic input low voltage	V_{IL2}			0.8	V
Logic output low voltage	V_{OL}	I_{OL} = 16 mA		0.4	V
Logic output high voltage	V_{OH}	I_{OH} = -250 µA	2.4		V
Power supply current	I_{CC}	Outputs open; V_{CC} = 5.5 V; T_C = -55°C		127	mA
Output leakge current low, open drain outputs	I_{LOL}	V_{IN} = 0.4 V	-10	+10	µA
Output leakge current high, open drain outputs	I_{LOH}	V_{IN} = 2.4 V	-10	+10	µA
Input low current (input and bidirectional)	I_{IL}	V_{IN} = 0.4 V	-10	+10	µA
Input high current (input and bidirectional)	I_{IH}	V_{IN} = 2.4 V	-10	+10	µA
Maximum frequency 01 circuit	f_{MAX}	$4.5 V \leq V_{CC} \leq 5.5 V$;	4.0		MHz
02 circuit		C_L = 50 pF ±10%	2.5		MHz
Clock input capacitance	C_{I1}	T_C = +25°C		20	pF
Logic input capacitance	C_{I2}			5	pF
Output and bidirectional capacitance	C_O			10	pF

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TABLE I. Electrical performance characteristics.

Parameter	Symbol	Conditions $-55°C \le T_C \le 125°C$; $V_{CC} = 5.0\ V \pm 10\%$ unless otherwise specified	Figure no. 1/	Reference no. 2/	Device types				Unit
					01		02		
					Min	Max	Min	Max	
Clock cycle time	t_{cyc1}	C_L = 50 pF ±10%	8	1	250	2,000	400	2,000	ns
CLK/TRG cycle time	t_{cyc2}		"	21	(2tcC1)		(2tcC1)		ns
Clock time Rise	t_{rC1}		"	5		20		30	ns
Fall	t_{fC1}		"	4		20		30	ns
CLK/TRG time Rise	t_{rC2}		"	22		50		50	ns
Fall	t_{fC2}		"	23		50		50	ns
Clock width High	t_{PWH1} 3/		"	2	105	2,000	170	2,000	ns
Low	t_{PWL1} 4/		"	3	105	2,000	170	2,000	ns
CLK/TRG width High	t_{PWH2} 5/		"	25	200		200		ns
Low	t_{PWL2} 6/		"	24	200		200		ns
CS0, CS1 to clock ↑ setup	t_{SLH1}		"	7	160		250		ns
	t_{SHL1}		"	7	160		250		ns
$\overline{CE}$ to clock ↑ setup	t_{SLH2}		"	8	150		200		ns
$\overline{IORQ}$ ↓ to clock ↑ setup	t_{SHL3}		"	9	115		250		ns
$\overline{RD}$ ↓ to clock ↑ setup	t_{SHL4}		"	10	115		240		ns
Data in to clock ↑ setup	t_{SLH5}		"	13	50		60		ns
	t_{SHL5}		"	13	50		60		ns
$\overline{M1}$ ↓ to clock ↑ setup	t_{SHL6}		"	14	90		210		ns
CLK/TRG ↑ to clock ↑ (time for enabling of pre-scaler on following clock ↑ timer mode)	t_{SLH7}		"	27	210		210		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Parameter	Symbol	Conditions $-55°C < T_C < 125°C$; $V_{CC} = 5.0\ V \pm 10\%$ unless otherwise specified	Figure no. 1/	Reference no. 2/	Device types 01 Min	01 Max	02 Min	02 Max	Unit
CLK/TRG ↑ to clock ↑ (time for immediate count counter mode)	t_{SLH8}	C_L = 50 pF ±10%	8	26	210		300		ns
All hold times	t_{WHL1}		"	6	0		0		ns
	t_{WLH1}		"	6	0		0		ns
Clock ↓ to data out delay	t_{PLH1}		"	11		200		240	ns
	t_{PHL1}		"	11		200		240	ns
Clock ↓ to data out float delay	t_{PLZ1}		"	12		110		230	ns
	t_{PHZ1}		"	12		110		230	ns
$\overline{MI}$ ↓ to IEO ↓ delay (Interrupt immediately preceeding $\overline{MI}$)	t_{PHL2}		"	15		190		300	ns
$\overline{IORQ}$ ↓ to data out delay (INTA cycle)	t_{PLH3}		"	16		160		340	ns
	t_{PHL3}		"	16		160		340	ns
IEI ↓ to IEO ↓ delay	t_{PHL4}		"	17		130		190	ns
IEI ↑ to IEO ↑ delay (after ED decode)	t_{PLH5}		"	18		160		220	ns
Clock ↓ to $\overline{INT}$ ↓ delay (timer mode)	t_{PHL6} 7/		"	19		(t_{cyc1} +140)		(t_{cyc1} +200)	ns
CLK/TRG ↓ to $\overline{INT}$ ↓ (TsCTRr(Cr) satisfied)	t_{PHL7} 7/		"	20		(t_{cyc1} +150)		(t_{cyc1} +230)	ns
(TsCTRr(Cr) not satisfied) counter mode	t_{PHL2} 7/		"	30		($2T_{cyc1}$ +370)		($2T_{cyc1}$ +530)	ns
Clock ↑ to ZC/TO ↑ delay	t_{PLH9}		"	28		190		260	ns
Clock ↑ to ZC/TO ↓ delay	t_{PHL10}		"	29		190		190	ns

1/ Figure 8 represents the electrical performance characteristics associated with the ac limits in table I.
2/ The reference number refers to the position where the parameter being tested appears on figure 8.
3/ $t_{PWH1} = t_{cyc1} - t_{rc1} - t_{fc1} - t_{PWL1}$.
4/ $t_{PWL1} = t_{cyc1} - t_{rc1} - t_{fc1} - t_{PWH1}$.
5/ $t_{PWH2} = t_{cyc2} - t_{rc2} - t_{fc2} - t_{PWL2}$.
6/ $t_{PWL2} = t_{cyc2} - t_{rc2} - t_{fc2} - t_{PWH2}$.
7/ Use load in figure 7, all other terminals use load in figure 6.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and on figure 8, and unless otherwise specified, apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III along with table IV and figure 8 which constitute the minimum electrical test requirements for screening, and quality conformance by device class are specified in table II.

3.6 Register definition. The register definitions shall be as specified in table V.

3.7 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 107 (see MIL-M-38510, appendix E).

3.9 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.10 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III unless otherwise specified)
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1, 7
Final electrical test parameters (method 5004)	*1, 2, 3, *7, 8, **9, 10, 11
Group A test requirements (method 5005)	1, 2, 3, 4, 7, 8, **9, 10, 11, 12
Group C end-point electrical parameters (method 5005)	1, 2, 3, 7, 8
Group D end-point electrical parameters (method 5005)	1, 2, 3, 7, 8

*PDA applies to subgroups 1 and 7, (see 4.2c).
**Subgroup 9 is run concurrently with subgroup 7 with a combined LTPD of 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4 or equivalent.

(2) T_A = 125°C, minimum.

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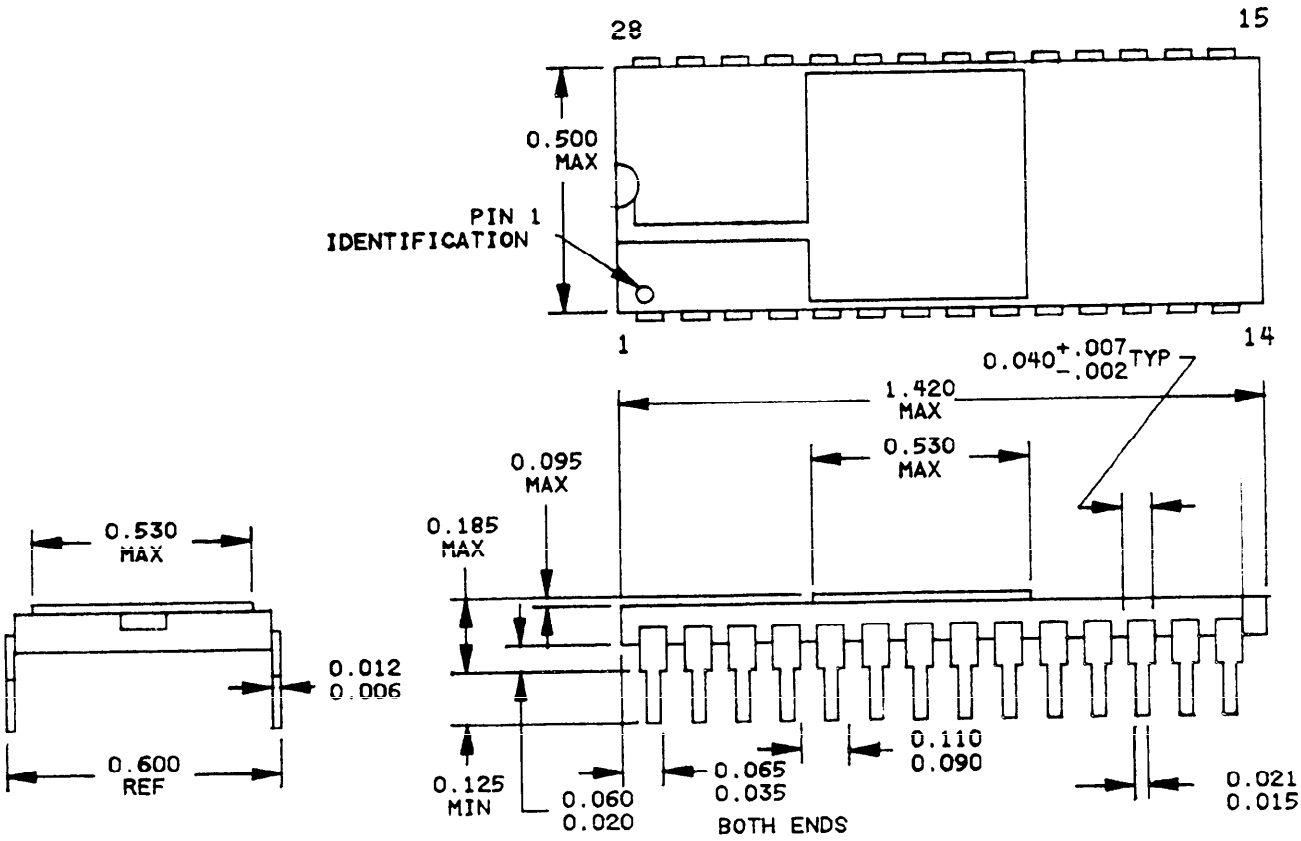


FIGURE 1. Case outline (28-lead, 9/16" x 1-7/16" dual-in-line package.

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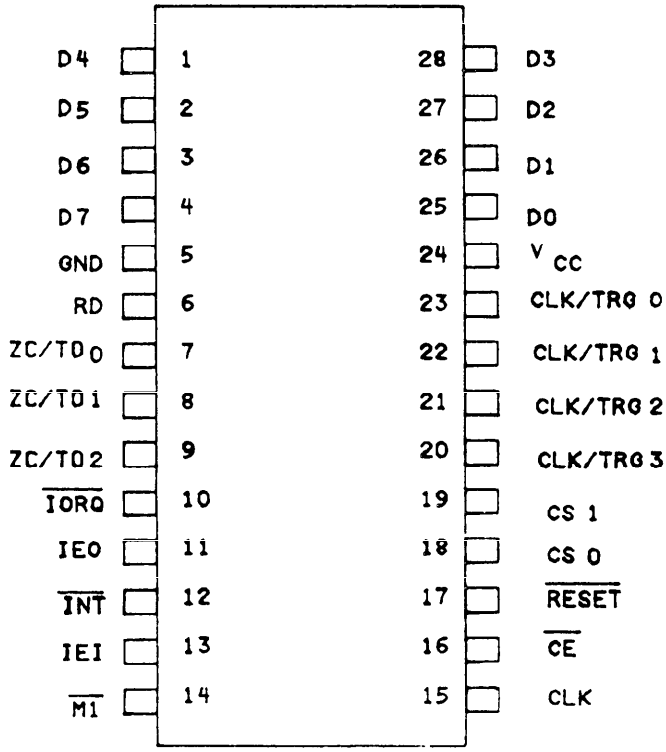


FIGURE 2. Terminal connections.

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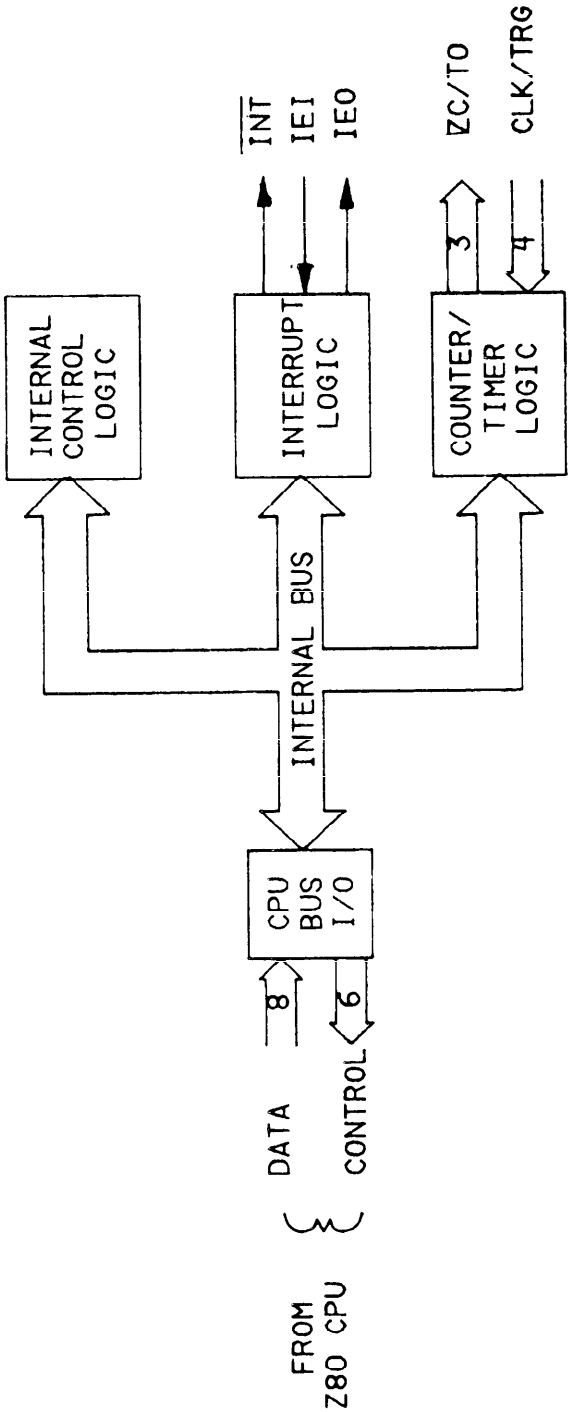
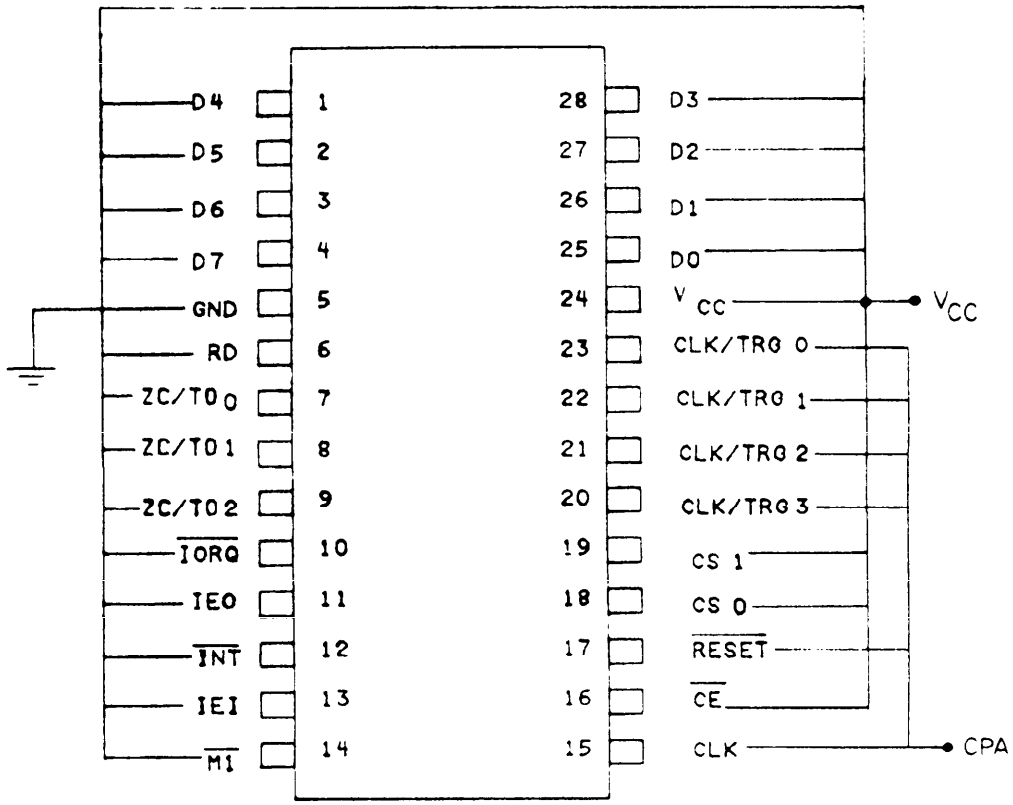


FIGURE 3. Functional block diagram.

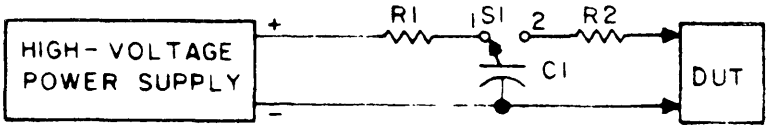
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NOTES:
1. CPA = 100 kHz ±10% square wave.
   50% duty cycle
2. V_{CC} shall be no less than 4.5 V at device terminal.
   V_{CC} shall be no more than 5.5 V at device terminal.
3. V_{ILC} shall be no more than 0.5 V at CPA.
   V_{IHC} shall be no less than 3.9 V at CPA.

FIGURE 4. Burn-in and life test circuit.

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NOTES:
1. V_{ZAP} = 400 V measured at the device input terminal.
2. 10 MΩ $\leq$ R1 $\leq$ 50 MΩ.
3. R2 = 1.5 kΩ.
4. C1 = 100 pF.
5. S1 = Hg-wetted "bounceless" relay break before make type.
6. All pins of D.U.T. not associated with V_{ZAP} test shall be open (NC).

FIGURE 5. High voltage (V_{ZAP}) test circuit.

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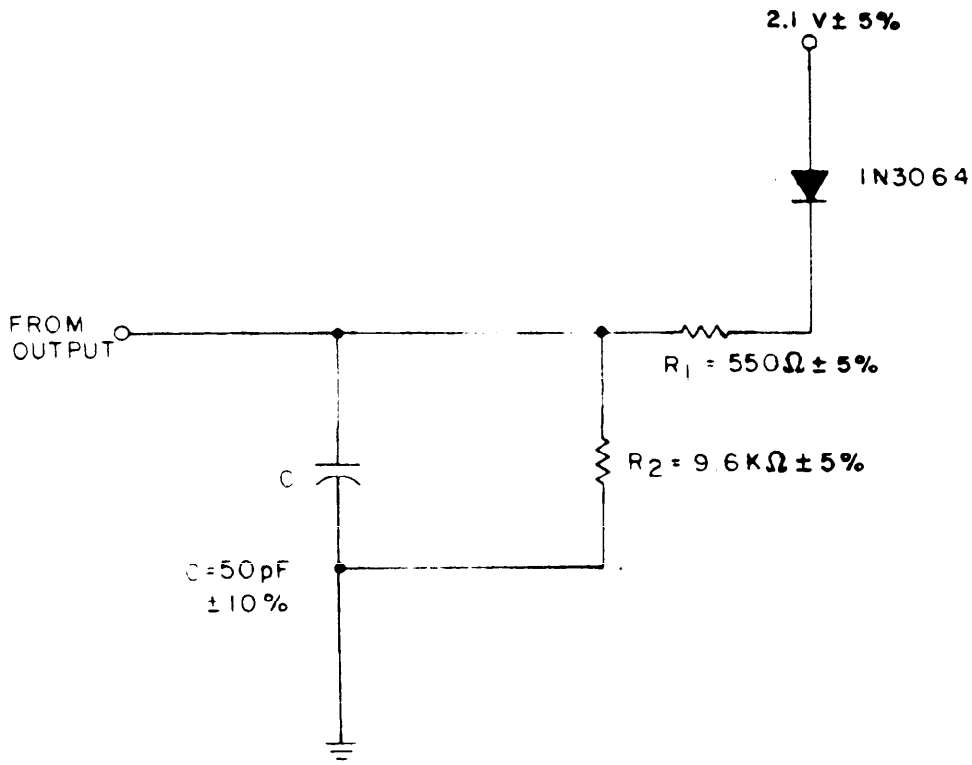
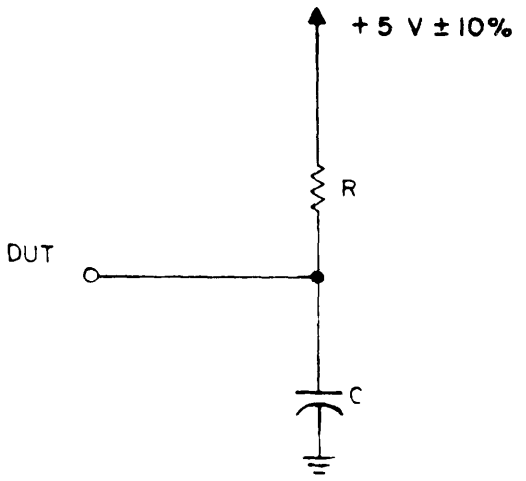


FIGURE 6. Operating load circuit for all standard outputs for ac and functional testing.

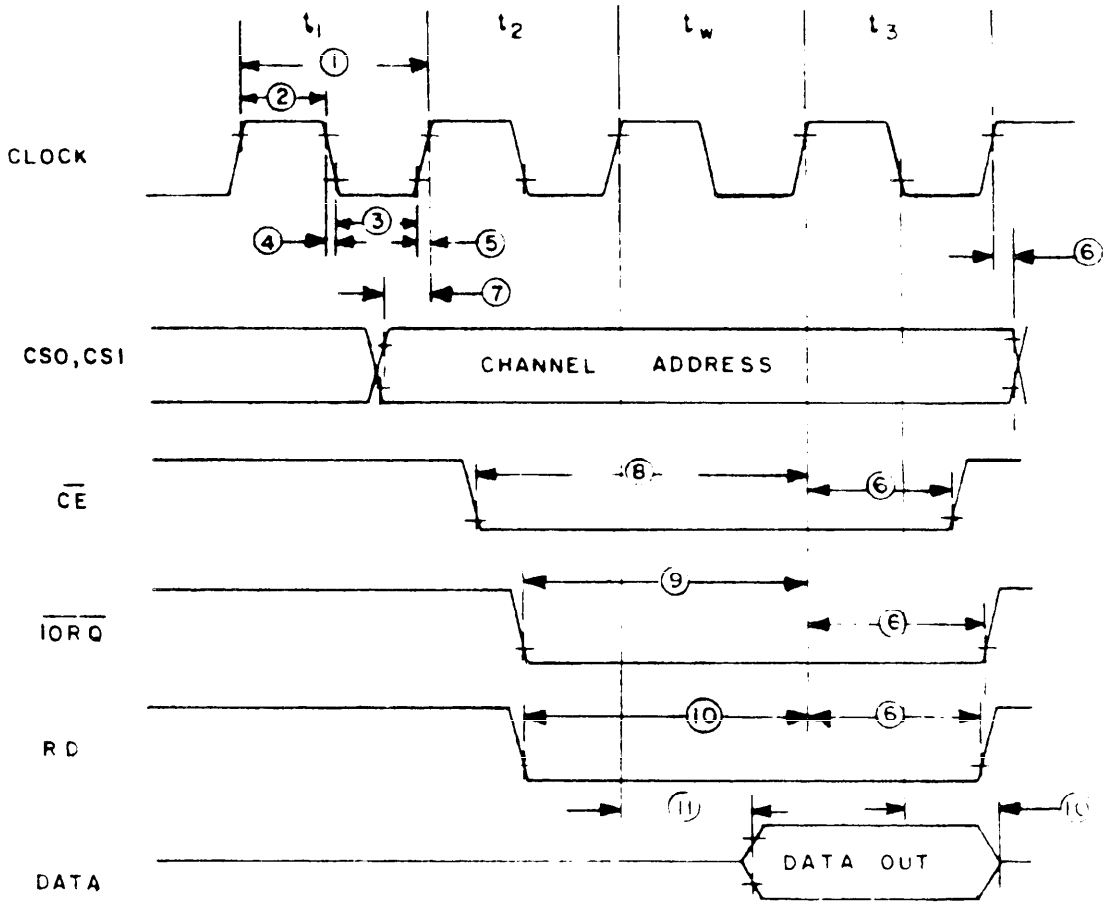
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NOTES:
1. C = 50 pF ±5% including tester, wiring and stray capacitance.
2. R = 2.2 kΩ ±5%.

FIGURE 7. Output load circuit for open drain outputs for ac and functional testing.

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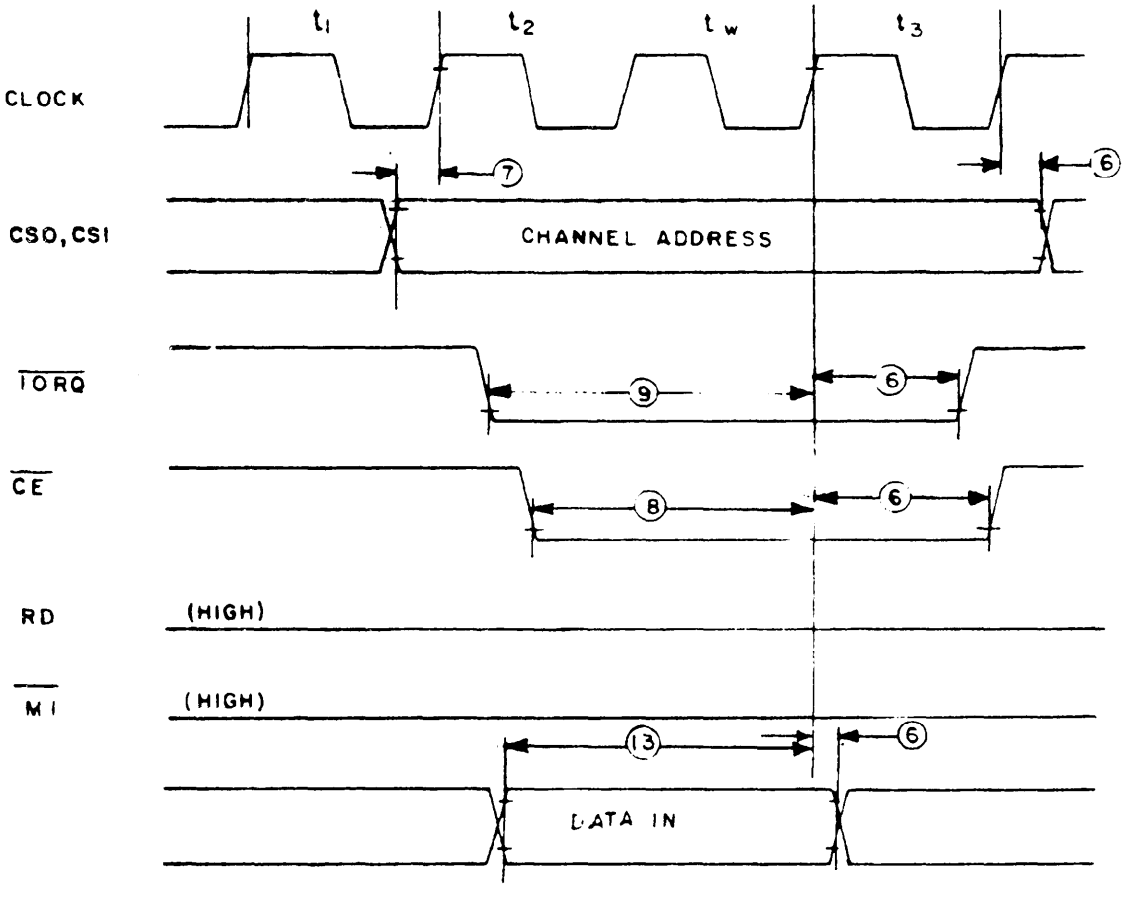


NOTE: No additional wait states are allowed.

READ CYCLE

FIGURE 8. Timing diagram.

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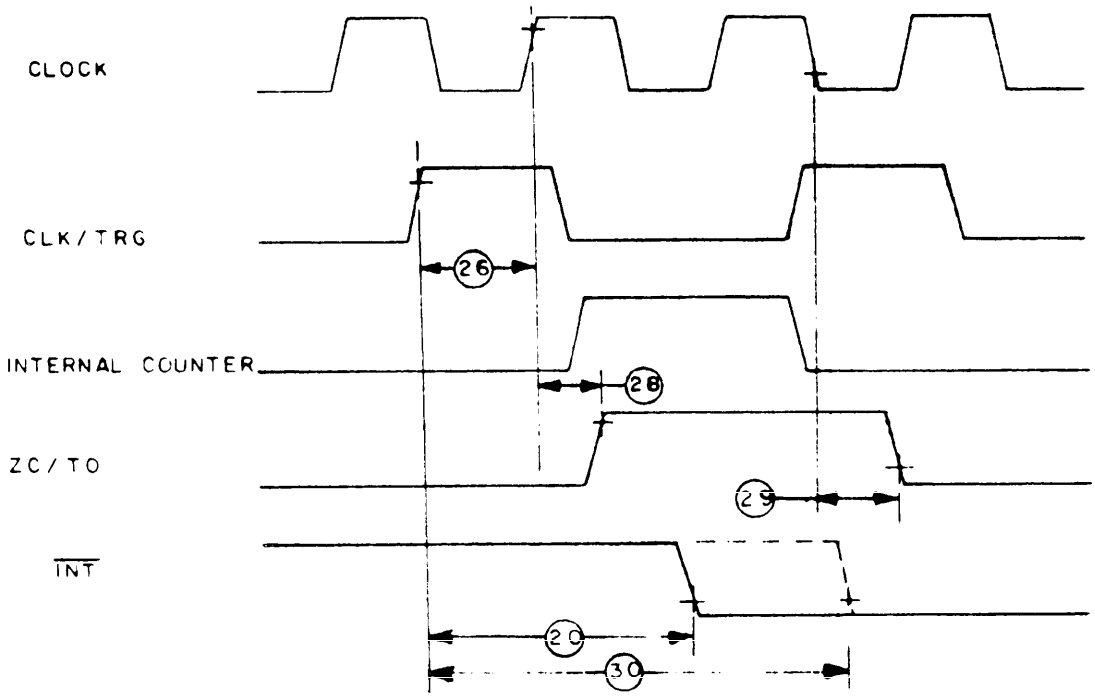


NOTE: No additional wait states are allowed.

WRITE CYCLE

FIGURE 8. Timing diagram - Continued.

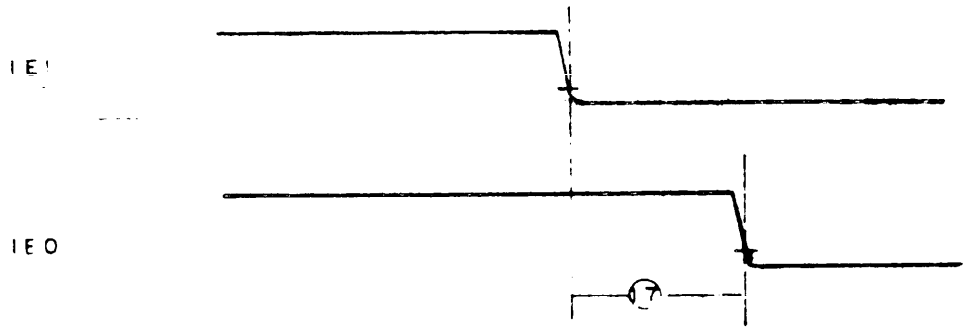
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COUNTER MODE TIMING

FIGURE 8. Timing diagram - Continued.

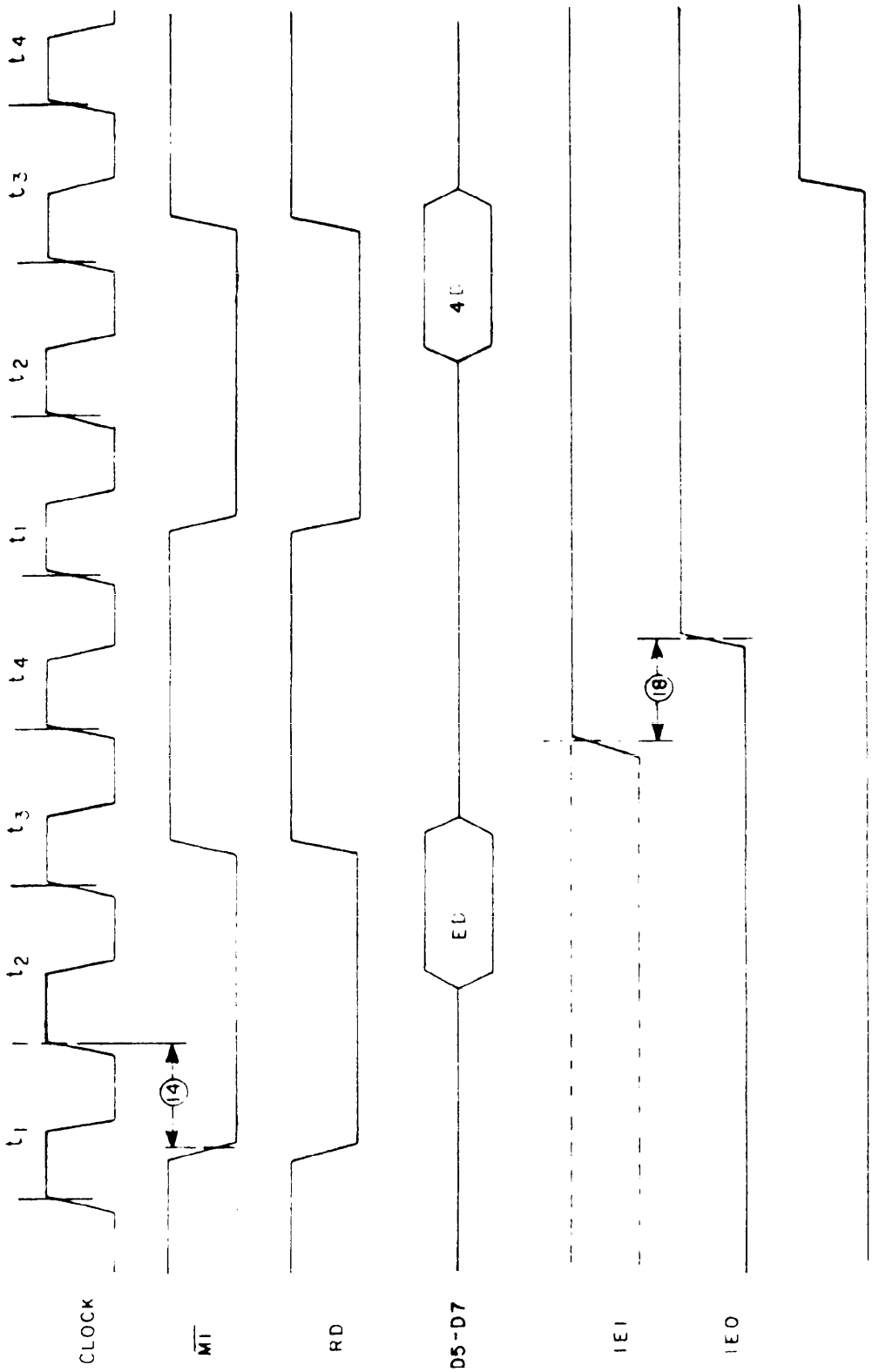
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IEI/IEO TIMING

FIGURE 8. Timing diagram - Continued.

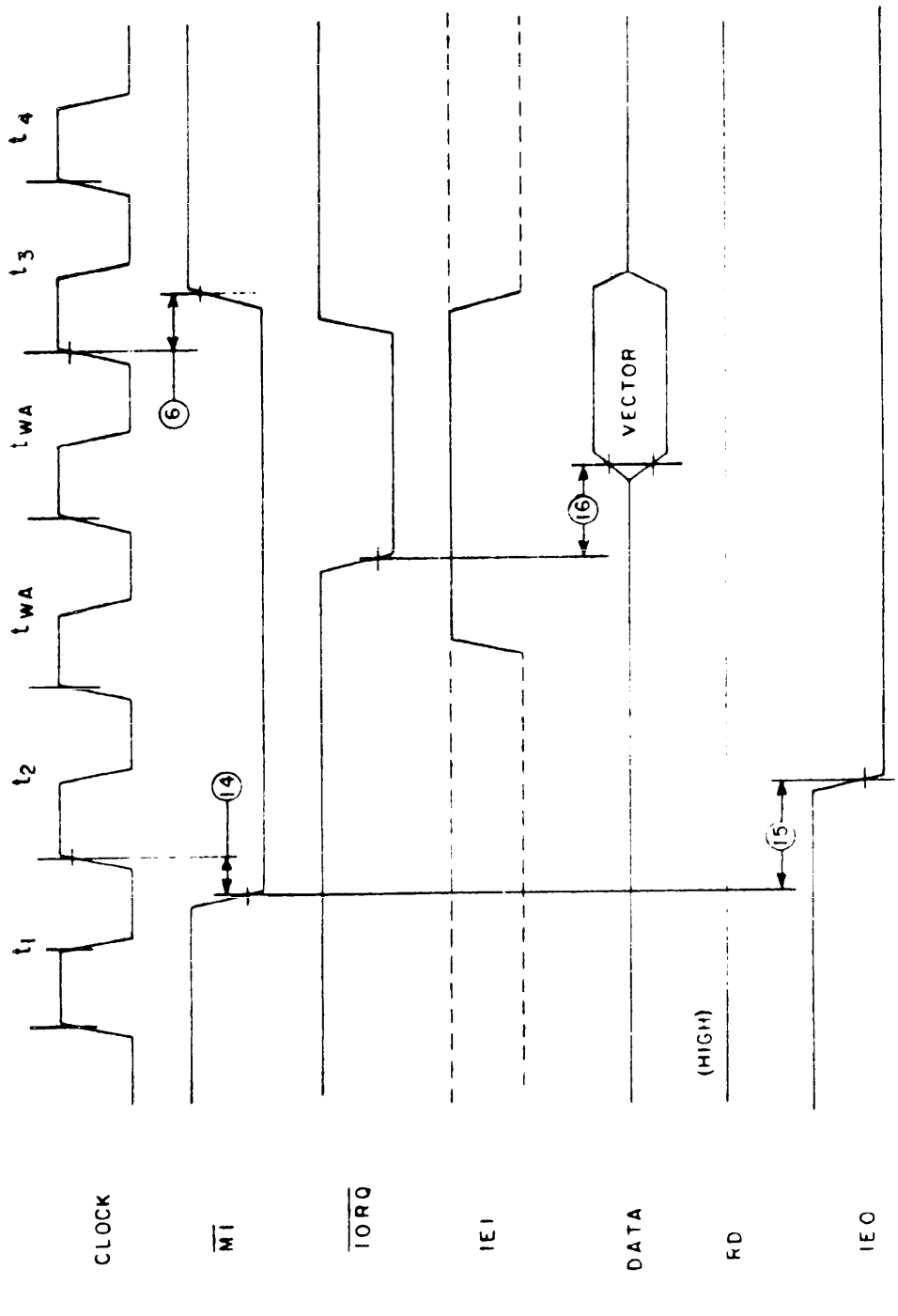
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RETURN FROM INTERRUPT TIMING

FIGURE 8. Timing diagram - Continued.

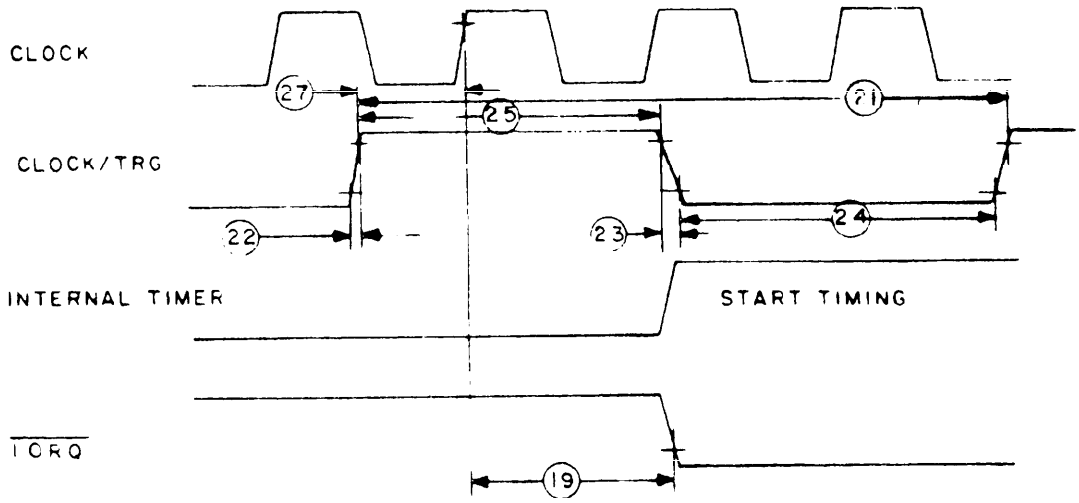
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NOTE: t_{wa} = Automatic wait state. INTERRUPT ACKNOWLEDGE TIMING.

FIGURE 8. Timing diagram - Continued.

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TIMER MODE TIMING

FIGURE 8. Timing diagram - Continued.

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TABLE III. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated are open)

Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	Measured terminal	Limits Min	Limits Max	Unit
			D4	D5	D6	D7	GND	RD	WC/TG3	ZC/TG1	ZC/TG2	CNT		INT	TG1	WT	CLK	CE	RESET	CS2	CS1	CLR/TRG1	CLR/TRG2	CLR/TRG3	CLR/TRG4	V_{CC}	D0	D1	D2	D3				
T_C = 25°C																																		
V_{CC}	3006	1					GND	0.8 V				0.8 V			0.8 V	0.8 V	4.9 V	2.0 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	5.5 V					28		32	mA
I_{CCT}		2					"	"				"		0.4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		12	10	15	μA
I_{CCH}		3					"	"				"		2.4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		12	"	"	"
I_{IH}		4	2.4 V				"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		1	"	"	"
		5		2.4 V			"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		2	"	"	"
		6			2.4 V		"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		3	"	"	"
		7				2.4 V	"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		4	"	"	"
		8					"	2.4 V				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		6	"	"	"
		9					"	0.8 V				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		10	"	"	"
		10					"	"				2.4 V		"	2.4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"		13	"	"	"
		11					"	"				0.8 V		"	0.8 V	2.4 V	2.4 V	"	"	"	"	"	"	"	"	"	"	"	"		14	"	"	"
		12					"	"				"		"	"	0.8 V	4.9 V	2.4 V	"	"	"	"	"	"	"	"	"	"	"		15	"	"	"
		13					"	"				"		"	"	"	"	2.3 V	"	"	"	"	"	"	"	"	"	"	"		16	"	"	"
		14					"	"				"		"	"	"	"	"	2.4 V	"	"	"	"	"	"	"	"	"	"		17	"	"	"
		15					"	"				"		"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"		18	"	"	"
		16					"	"				"		"	"	"	"	"	"	2.4 V	"	"	"	"	"	"	"	"	"		19	"	"	"
		17					"	"				"		"	"	"	"	"	"	0.8 V	2.4 V	"	"	"	"	"	"	"	"		20	"	"	"
		18					"	"				"		"	"	"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"		21	"	"	"
		19					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		22	"	"	"
		20					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		23	"	"	"
		21					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		25	"	"	"
		22					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		26	"	"	"
		23					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		27	"	"	"
		24					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		28	"	"	"
		25	0.4 V				"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		1	"	"	"
		26		0.4 V			"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		2	"	"	"
		27			0.4 V		"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		3	"	"	"
		28				0.4 V	"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		4	"	"	"
		29					"	0.4 V				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		6	"	"	"
		30					"	0.8 V				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		10	"	"	"
		31					"	"				0.4 V		"	0.4 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"		13	"	"	"
		32					"	"				0.8 V		"	0.8 V	0.4 V	0.4 V	"	"	"	"	"	"	"	"	"	"	"	"		14	"	"	"
		33					"	"				"		"	"	0.8 V	4.9 V	0.4 V	"	"	"	"	"	"	"	"	"	"	"		15	"	"	"
		34					"	"				"		"	"	"	"	2.0 V	"	"	"	"	"	"	"	"	"	"	"		16	"	"	"
		35					"	"				"		"	"	"	"	"	0.4 V	"	"	"	"	"	"	"	"	"	"		17	"	"	"
		36					"	"				"		"	"	"	"	"	0.8 V	"	"	"	"	"	"	"	"	"	"		18	"	"	"
		37					"	"				"		"	"	"	"	"	"	0.4 V	0.4 V	"	"	"	"	"	"	"	"		19	"	"	"
		38					"	"				"		"	"	"	"	"	"	"	0.8 V	0.4 V	"	"	"	"	"	"	"		20	"	"	"
		39					"	"				"		"	"	"	"	"	"	"	"	0.8 V	0.4 V	"	"	"	"	"	"		21	"	"	"
		40					"	"				"		"	"	"	"	"	"	"	"	"	0.8 V	0.4 V	"	"	"	"	"		22	"	"	"
		41					"	"				"		"	"	"	"	"	"	"	"	"	"	0.8 V	0.4 V	"	"	"	"		23	"	"	"
		42					"	"				"		"	"	"	"	"	"	"	"	"	"	"	0.8 V	"	"	"	"		25	"	"	"
		43					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		26	"	"	"
		44					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		27	"	"	"
		45					"	"				"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		28	"	"	"
2 T_C = -125°C	Repeat tests 1 through 45 with the limit for test 1 changed to 91 mA																																	
3 T_C = -55°C	Repeat tests 1 through 45 with the limit for test 1 changed to 127 mA																																	

See footnotes at end of table.

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TABLE I-II. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pin not designated are open).

Subgroup	Symbol	MIL-STD-883 method	Test no	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	Measured terminal	Test limits Min	Test limits Max	Unit
				I4	I5	D6	D7	GND	RD	IZC/TOO	IZC/TO1	ZC/TO2	TOZC	IE3	INT	IE1	WT	CLK	CE	RESET	CS0	CS1	CLE/IRG3	CLR/IRG2	CLR/IRG1	CLR/IRQ0	V_{CC}	D0	D1	D2	D3				
4	C_{11}		46	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	1/	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	15		pF	
	C_{12}		47	-	-	-	-	-	1/ GND	-	-	-	1/ GND	-	-	-	-	GND	-	-	-	-	-	-	-	-	-	-	-	-	6			
			48	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	10			
			49	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	13			
			50	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	14			
			51	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	1/ GND	1/ GND	-	-	-	-	-	-	-	-	-	-	-	15			
			52	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	16			
			53	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	17			
			54	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	18			
			55	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	19			
			56	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	20			
			57	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	21			
			58	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	22			
			59	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/	-	-	-	-	23			
	C_{13}		60	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	GND	-	-	-	-	1	10		
			61	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2			
			62	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3			
			63	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4			
			64	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	7			
			65	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8			
			66	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	9			
			67	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	11			
			68	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	12			
			69	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	-	24			
			70	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	-	27			
			71	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1/ GND	28			
			72	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
7 T_C = +25°C	2/,3/, 5/	3014																															
8	2/,3/, 5/	3014	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125°C.																														
9 T_C = +25°C	2/,3/, 4/	3014																															
10 T_C = +125°C	2/,3/, 4/	3014																															
11 T_C = -55°C	2/,3/, 4/	3014																															

See footnotes at end of table

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TABLE III. Group A inspection for device types 01 and 02.
Subgroup 12 limits +25°C, +125°C

Test number	Symbol	Terminal(s)	Figure number	Reference number	Device type 01		Device type 02		Unit
					Minimum	Maximum	Minimum	Maximum	
73	t_{PLH1} t_{PHL1}	D0-D7	8	11		200		240	ns
74	t_{PLZ1} t_{PHZ1}	"	"	12		110		230	"
75	t_{PHL2}	$\overline{M1}$	"	15		190		300	"
76	t_{PLH3} t_{PHL3}	D0-D7	"	16		160		340	"
77	t_{PHL4}	IEI	"	17		130		190	"
78	t_{PLH5}	IEO	"	18		160		220	"
79	t_{PHL6}	$\overline{INT}$	"	19		t_{cyc1}+140		t_{cyc1}+200	"
80	t_{PHL7}	$\overline{INT}$	"	20		t_{cyc1}+160		t_{cyc1}+230	"
81	t_{PHL8}	$\overline{INT}$	"	30		t_{cyc1}+370		t_{cyc1}+530	"
82	t_{PLH9}	ZC/TO	"	28		190		260	"
83	t_{PLH10}	ZC/TO	"	29		190		190	"

1/ The capacitance measurements shall be made between the indicated terminal and ground at a frequency of 1 MHz. The dc bias of the measuring instrument shall be less than ±0.1 V. The ac signal amplitude shall be less than 50 mV rms.

2/ For functional tests, input/output conditions, A, B, C, and D appear in table IV.

3/ Table VI functional tests shall be repeated for input/output conditions A through D of table IV.

4/ All subgroups 9, 10, and 11 tests are run concurrently with subgroups 7 and 8.

5/ V_{OH} and V_{OL} are tested as part of subgroups 7 and 8.

6/ Where propagation delay tests for subgroup 12 indicate a group of pins and where two or more propagation delays are specified for a test, only the reading for the pin with the worst case delay requires recording. In some cases this will be a minimum reading while for others it will be a maximum as indicated in the table.

7/ Table VI functional tests shall be repeated for input/output conditions A through D of table IV for tests 73 through 83 of table III.

8/ When patterns are applied as specified for subgroups 7 and 8, they verify the dynamic, functional, output threshold, and switching parameters at T_C = +25°C, -55°C, and +125°C.

9/ Figure 8 represents the timing diagrams associated with subgroups 9, 10, and 11.

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TABLE IV. Input and output conditions for table III for device type 01.

Symbol	Reference (table I)	Terminal(s)	A	B	C	D	Unit
V_{CC}		V_{CC}	5.5	4.5	4.5	5.5	V
V_{IH1}		CLK	4.9	3.9	3.9	4.9	V
V_{IL1}		CLK	0.45	0.45	0.45	0.45	V
V_{IH2}		All logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}		All logic inputs	0.8	0.8	0.8	0. 8	V
V_{OH}		All logic outputs	2.4	2.4	2.4	2.4	V
V_{OL}		All logic outputs	0.4	0.4	0.4	0.4	V
t_{cyc1}	1	CLK	250	250	2,000	2,000	ns
t_{PWH1}	2	CLK	105	105	105		ns
t_{PWL1}	3	CLK				105	ns
t_{rc1}	5	CLK	20	20	20	20	ns
t_{fc1}	4	CLK	20	20	20	20	ns
t_{cyc2}	21	CLK/TRG inputs	500	500	4,000	4,000	ns
t_{PWH2}	25	CLK/TRG inputs	200	200	200		ns
t_{PWL2}	24	CLK/TRG inputs				200	ns
t_{rc2}	22	CLK/TRG inputs	50	50	50	50	ns
t_{fc2}	23	CLK/TRG inputs	50	50	50	50	ns
t_{SLH1} t_{SHL1}	7	CS0, CS1	160	160	160	160	ns
t_{SLH2}	8	$\overline{CE}$	150	150	150	150	ns
t_{SHL3}	9	$\overline{IORQ}$	115	115	115	115	ns
t_{SHL4}	10	$\overline{RD}$	115	115	115	115	ns

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TABLE IV. Input and output conditions for table III for device type 01 - Continued.

Symbol	Reference (table I)	Terminal(s)	A	B	C	D	Unit
t_{SHL5} t_{SLH5}	13	D0-D7	50	50	50	50	ns
t_{SHL6}	14	$\overline{MT}$	90	90	90	90	ns
t_{SLH7}	27	CLK/TRG	210	210	210	210	ns
t_{SLH8}	26	CLK/TRG	210	210	210	210	ns
t_{HHL1} t_{HLH1}	6	All logic inputs	0	0	0	0	ns
t_{PLZ1} t_{PHZ1}	12	D0-D7	110	110	110	110	ns
t_{PLH1} t_{PHL1}	11	D0-D7	200	200	200	200	ns
t_{PHL2}	15	IEO	190	190	190	190	ns
t_{PLH3} t_{PHL3}	16	D0-D7	160	160	160	160	ns
t_{PHL4}	17	IEO	130	130	130	130	ns
t_{PLH5}	18	IEO	160	160	160	160	ns
t_{PHL6}	19	$\overline{INT}$	390	390	4,140	4,140	ns
t_{PHL7}	20	$\overline{INT}$	410	410	4,160	4,160	ns
t_{PLH8}	30	$\overline{INT}$	870	870	4,370	4,370	ns
t_{PLH9}	28	ZC/TO outputs	190	190	190	190	ns
t_{PLH10}	29	ZC/TO outputs	190	190	190	190	ns

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TABLE IV. Input and output conditions for table III for device type 02.

Symbol	Reference (table I)	Terminal(s)	A	B	C	D	Unit
V_{CC}		V_{CC}	5.5	4.5	4.5	5.5	V
V_{IH1}		CLK	4.9	3.9	3.9	4.9	V
V_{IL1}		CLK	0.45	0.45	0.45	0.45	V
V_{IH2}		All logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}		All logic inputs	0.8	0.8	0.8	0. 8	V
V_{OH}		All logic outputs	2.4	2.4	2.4	2.4	V
V_{OL}		All logic outputs	0.4	0.4	0.4	0.4	V
t_{cyc1}	1	CLK	400	400	2,000	2,000	ns
t_{PWH1}	2	CLK	170	170	170		ns
t_{PWL1}	3	CLK				170	ns
t_{rc1}	5	CLK	30	30	30	30	ns
t_{fc1}	4	CLK	30	30	30	30	ns
t_{cyc2}	21	CLK/TRG inputs	800	800	4,000	4,000	ns
t_{PWH2}	25	CLK/TRG inputs	200	200	200		ns
t_{PWL2}	24	CLK/TRG inputs				200	ns
t_{rc2}	22	CLK/TRG inputs	50	50	50	50	ns
t_{fc2}	23	CLK/TRG inputs	50	50	50	50	ns
t_{SLH1} t_{SHL1}	7	CS0, CS1	250	250	250	250	ns
t_{SLH2}	8	$\overline{CE}$	200	200	200	200	ns
t_{SHL3}	9	$\overline{IORQ}$	250	250	250	250	ns
t_{SHL4}	10	$\overline{RD}$	240	240	240	240	ns

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TABLE IV. Input and output conditions for table III for device type 02 - Continued.

Symbol	Reference (table I)	Terminal(s)	A	B	C	D	Unit
t_{SHL5}	13	D0-D7	60	60	60	60	ns
t_{SHL6}	14	$\overline{MT}$	210	210	210	210	ns
t_{SLH7}	27	CLK/TRG	210	210	210	210	ns
t_{SLH8}	26	CLK/TRG	300	300	300	300	ns
t_{HHL1} t_{HLH1}	6	All logic inputs	0	0	0	0	ns
t_{PLZ1} t_{PHZ1}	12	D0-D7	230	230	230	230	ns
t_{PLH1} t_{PHL1}	11	D0-D7	240	240	240	240	ns
t_{PHL2}	15	IEO	300	300	300	300	ns
t_{PLH3} t_{PHL3}	16	D0-D7	340	340	340	340	ns
t_{PHL4}	17	IEO	190	190	190	190	ns
t_{PLH5}	18	IEO	220	220	220	220	ns
t_{PHL6}	19	$\overline{INT}$	600	600	4,200	4,200	ns
t_{PHL7}	20	$\overline{INT}$	630	630	4,230	4,230	ns
t_{PLH8}	30	$\overline{INT}$	1,330	1,330	4,530	4,530	ns
t_{PLH9}	28	ZC/TO outputs	260	260	260	260	ns
t_{PLH10}	29	ZC/TO outputs	190	190	190	190	ns

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TABLE V. Register definitions.

Programming

Each Z-80 CTC channel must be programmed prior to operation. Programming consists of writing two words to the IO port that corresponds to the desired channel. The first word is a control word that selects the operating mode and other parameters; the second word is a time constant, which is a binary data word with a value from 1 to 256. A time constant word must be preceded by a channel control word.

After initialization, channels may be reprogrammed at any time. If updated control and time constant words are written to a channel during the count operation, the count continues to zero before the new time constant is loaded into the counter.

If the interrupt on any Z-80 CTC channel is enabled, the programming procedure should also include an interrupt vector. Only one vector is required for all four channels, because the interrupt logic automatically modifies the vector for the channel requesting service.

A control word is identified by a 1 in bit 0. A 0 in bit 2 indicates a time constant word is to follow. Interrupt vectors are always addressed to Channel 0, and identified by a 0 in bit 0.

Addressing

During programming, channels are addressed with the channel select pins CS1 and CS2. A 2-bit binary code selects the appropriate channel as shown in the following.

Channel	CS1	CS0
0	0	0
1	0	1
2	1	0
3	1	1

Reset

The CTC has both hardware and software resets. The hardware reset terminates all down-counts and disables all CTC interrupts by resetting the interrupt bits in the control registers. In addition, the ZC/TO and interrupt outputs go inactive, IEO reflects IEI, and DO-D7 go to the high-impedance state. All channels must be completely reprogrammed after a hardware reset.

The software reset is controlled by bit 1 in the channel control word. When a channel receives a software reset, it stops counting. When a software reset is used, the other bits in the control word also change the contents of the channel control register. After a software reset a new time constant word must be written to the same channel.

If the channel control word has both bits D1 and D2 set to 1, the addressed channel stops operating, pending a new time constant word. The channel is ready to resume after the new constant is programmed. In timer mode, if D3 = 0, operation is triggered automatically when the time constant word is loaded.

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the contract.

TABLE V. Register definitions - Continued.

Channel control word programming	The channel control word is shown on figure 9. It sets the mode and parameters described below.

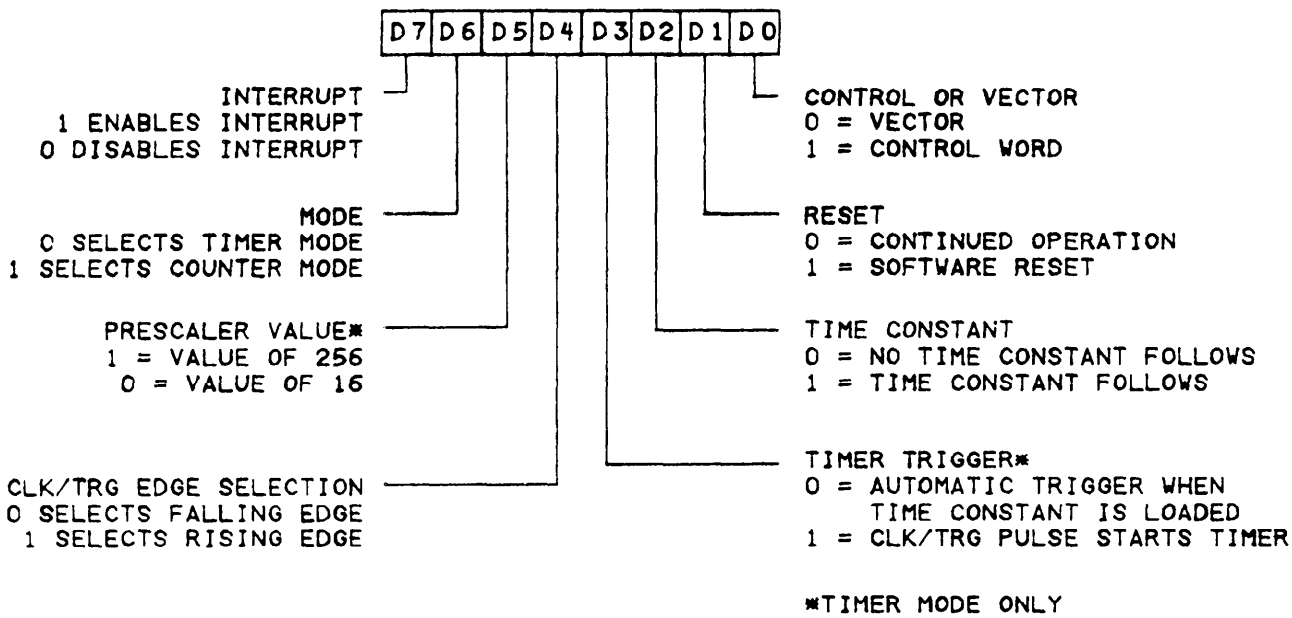


FIGURE 9. Channel control word.

Interrupt enable	D7 enables the interrupt, so that an interrupt output ($\overline{INT}$) is generated at zero count. Interrupts may be programmed in either mode and may be enabled or disabled at any time.
Operating mode	D6 selects either timer or counter mode.
Prescaler factor (timer mode only)	D5 selects factor - either 16 or 256.
Trigger slope	D4 selects the active edge or slope of the CLK/TRG input pulses. Note that reprogramming the CLK/TRG slope during operation is equivalent to issuing an active edge. If the trigger slope is changed by a control word update while a channel is pending operation in timer mode, the result is the same as a CLK/TRG pulse and the timer starts. Similarly, if the channel is in counter mode, the counter decrements.

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TABLE V. Register definitions - Continued.

Trigger mode (timer mode only)	D3 selects the trigger mode for timer operation. When D3 is reset to 0, the timer is transferred automatically. The time constant word is programmed during an I/O write operation, which takes one machine cycle. At the end of the write operation there is a setup delay of one clock period. The timer starts automatically (decrements) on the rising edge of the second clock pulse (T2) of the machine cycle following the write operation. Once started, the timer runs continuously. At zero count the timer reloads automatically and continues counting without interruption or delay until stopped by a reset. When D3 is set to 1, the timer is triggered externally through the CLK/TRG input. The time constant word is programmed during an I/O write operation, which takes one machine cycle. The timer is ready for operation on the rising edge of the second clock pulse (T2) of the following machine cycle. Note that the first timer decrement follows the active edge of the CLK/TRG pulse by a delay time of one clock cycle if a minimum setup time to the rising edge of clock is met. If this minimum is not met, the delay is extended by another clock period. Consequently, for immediate triggering, the CLK/TRG input must precede T2 by one clock cycle plus its minimum setup time. If the minimum time is not met, the timer will start on the third clock cycle (T3). Once started the timer operates continuously, without interruption or delay, until stopped by a reset.
Time constant to follow	A 1 in D2 indicates that the next word addressed to the selected channel is a time constant data word for the time constant register. The time constant word may be written at any time. A 0 in D2 indicates no time constant word is to follow. This is ordinarily used when the channel is already in operation and the new channel control word is an update. A channel will not operate without a time constant value. The only way to write a time constant value is to write a control word with D2 set.
Software reset	Setting D1 to 1 causes a software reset (see reset section).
Control word	Setting D0 to 1 identifies the word as a control word.

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TABLE V. Register definitions - Continued.

Time constant programming

Before a channel can start counting it must receive a time constant word from the CPU. During programming or reprogramming, a channel control word in which bit 2 is set must precede the time constant word to indicate that the next word is a time constant. The time constant word can be any value from 1 to 256 (figure 10). Note that 00_{16} is interpreted as 256.

In timer mode, the time interval is controlled by three factors:

a. The system clock period (ø)

b. The prescaler factor (P), which multiplies the interval by either 16 or 256

c. The time constant (T), which is programmed into the time constant register

Consequently, the time interval is the product of ø x P x T. The minimum timer resolution is 16 x ø (4 µs with a 4 MHz clock). The maximum timer interval is 256 x ø x 256 (16.4 ms with a 4 MHz clock). For longer intervals timers may be cascaded.

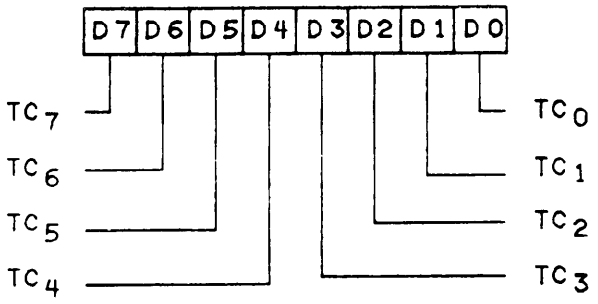


FIGURE 10. Time constant word.

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TABLE V. Register definitions - Continued.

Interrupt vector programming	If the Z-80 CTC has one or more interrupts enabled, it can supply interrupt vectors to the Z-80 CPU. To do so, the Z-80 CTC must be pre-programmed with the most significant five bits of the interrupt vector. Programming consists of writing a vector word to the IO port corresponding to the Z-80 CTC channel 0. Note that D0 of the vector word is always zero, to distinguish the vector from a channel control word. D1 and D2 are not used in programming the vector word. These bits are supplied by the interrupt logic to identify the channel requesting interrupt service with a unique interrupt vector (figure 11). Channel 0 has the highest priority.

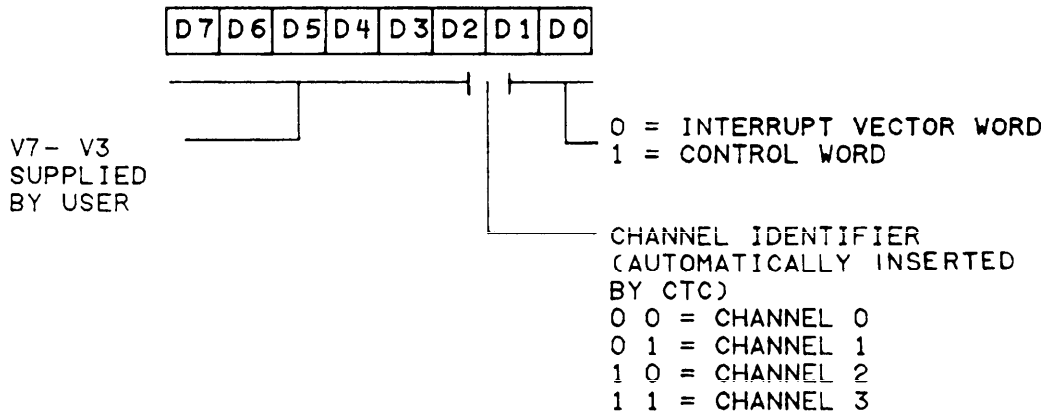


FIGURE 11. Interrupt vector word.

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b. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameter tests prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) is specified as 10 percent for class B devices based on failures from group A, subgroups 1 and 7 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7 after burn-in in that lot shall be used to determine the percent defective for the lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be perfomed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.3). Generic test data (see 6.8) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II, herein.

b. Subgroup 4 (C_{I1}, C_{I2}, and C_O measurements) shall be performed only after process or design changes which may affect device capacitance.

c. Subgroup 12 shall be added to group A inspection requirements for class B devices using an LTPD of 10 and consists of the procedures, test conditions and limits specified in table III. Variables data shall be taken for subgroup 12 only during initial verification and after process or design changes.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883. A special subgroup shall be added to the group B inspection requirements for class B. This subgroup shall consist of the tests, conditions, and limits specified in 4.5.2. The LTPD for this subgroup shall be 15.

4.4.3 Groups C and D inspections. Groups C and D inspections shall consist of the test subgroups and LTPD values shown in tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II, herein.

b. Steady state life test (method 1005 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4.

(2) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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4.5.2 High voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device number under test (DUT) shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 5.

a. Measure I_{IH} and I_{IL} at one input terminal of the device under test at 25°C. This measurement shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

b. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in two pulse sequence as follows and in accordance with the test circuit of figure 5:

(1) Input (-) to ground.

(2) Input (+) to ground.

c. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. After the V_{ZAP} test a DUT exhibiting leakage currents in excess of the specified limits is defective.

5. PACKAGING

5.1 Packaging requirements. .The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Test vector table. Table VI, which forms a part of this specification, is not printed herein because of its extreme length and complexity; and, even when presented as hard copy, it requires conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be obtained in a form compatible with test system architecture (magnetic tape usually preferred). The preparing activity, or Defense Electronics Supply Center as its agent, may be consulted for information on obtaining such output. For reference purposes, Rome Air Development Center (RADC-RBE-2) and Defense Electronics Supply Center (DESC-ECS) each maintain a limited number of dated, hard copy printouts of this table.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for failure analysis, (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These special requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by, or direct shipment to the government.

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6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal).
f_{MAX}	Maximum clock frequency.
V_{IN}	Voltage level at an input terminal.

6.5 Terms and definitions. The abbreviations, terms, symbols, and definitions used herein (including terms and symbols for device terminals) are defined in MIL-M-38510, MIL-STD-1331, and as follows:

D7-D0 — Data bus (bidirectional, 3-state). This bus is used to transfer all data and command words between the CPU and CTC. D0 is the least significant bit.

CS1-CS0 — Channel select (inputs active high). These pins form a two bit binary address which selects one of the four independent CTC channels for an I/O read or write.

	CS1	CS0
Channel 0	0	0
Channel 1	0	1
Channel 2	1	0
Channel 3	1	1

CLK — System clock (input). CLK is the single phase time base input.

$\overline{CE}$ — Chip enable (input, active low). A low level on this pin enables the CTC to accept control words, interrupt vectors, or time constant data from the data bus during an I/O write cycle, or to transmit the contents of a down counter to the CPU during an I/O read.

$\overline{M1}$ — Machine cycle one (input, active low). When M1 is active and the IORQ signal is active, the CTC is alerted to place an interrupt vector onto the data bus.

RD — Read (input, active low). When this line is inactive, and the CTC is enabled, control words, interrupt vectors, or time constant data can be written to the CTC. When active, the contents of any down counter can be read by the CPU without distrubing the count.

IEI — Interrupt enable in (input, active high). When this line is active, the CTC is able to interrupt the CPU.

IEO — Interrupt enable out (output, active high). This output is high only if IEI is high and the CPU is not servicing an interrupt from any CTC channel. In conjunction with IEI, this line can be used to implement a system-wide interrupt priority daisy chain.

$\overline{INT}$ — Interrupt request (output, open drain, active low). This signal becomes active whenever any CTC channel which has been programmed to enable interrupts has a zero count in its down counter.

$\overline{IORQ}$ — Input/output request (input, active low). Used with RD and CE, this input, when active, allows the transfer of data and control words between the CPU and CTC.

$\overline{RESET}$ — Reset (input, active low). This signal stops all channels from counting and resets the channel interrupt enable bits in all control registers. The INT output is forced to assume the inactive state, IEO reflects IEI, and the CTC's data bus output drivers go to their high impedance state.

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CLK/TRG3- CLK/TRG0	External clock/timer trigger (input, user selectable active high or low) In the counter mode, these inputs are the clocks of the four independent counters. In the timer mode, an active edge on these pins initiates the timing function. The user may select the active edge to be either the rising or falling edge.
ZC/TO2- ZC/TO0	Zero count/timeout (output, active high). In either counter or timer mode, when the down counter decrements to zero, an active high going pulse appears on these pins.

6.6 Logistic support. Lead material and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish "C" (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.7 Handling. MOS integrated circuits should be handled with certain precautions to avoid damage due to the accumulation of static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage; however, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

6.8 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the plant as the die represented. Group D (see 4.4.3) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.9 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types, and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	Z80A CTC
02	Z80 CTC

6.10 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensivenss of the changes.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

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